

Assignment 3 - common pitfalls

- Repeated union is not a new connection between components. Update sizes only after distinct roots merge.
- An MST minimises a forest's total cable cost; it does not provide shortest routes from a source.
- A disconnected graph needs a minimum spanning forest under this contract. Restart Prim at each component.
- SCC labels may differ between correct outputs. Deduplicate condensation edges; do not remove real prerequisite cycles by wishful thinking.
- DFS must distinguish active and finished vertices. A visited vertex alone does not prove a cycle.
- Relax DAG edges in topological order. Skip unreachable predecessors; null is different from distance -1.
- Use 64-bit costs and distances. Use explicit traversal stacks on deep graphs.